MEDICAL-ERP APPLICATION DEPLOYMENT ON SERVER

Application

1. B2B medical ERP system
2. client: MedPharma

Challenges

1. Inventory ---> Excel
2. data spread
3. Reporting ---> single point to get result

Solution

SAS application

1. ERP solution - medical
2. centralize data manage

Technical stack

1. Frontend - React
2. Backend - Springboot, java
3. Database - MongoDB Atlas

Team Size

1. 2 FD (frontend developer)
2. 2 BD (backend developer)
3. 1 T (Tester)
4. 1 DE (DevOps Engineer)
5. 1 PM (project manager)

Branching Strategy

1. Feature
2. Dev
3. Test
4. Uat
5. Prod
6. Hotfix

Application Link https://github.com/shubhamkalsait/edublitz-b2b-medical-erp.git

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Practical:

CONFIGURE MONGODB DATABASE

1. create a cluster (edublitz)
    1. click on the cluster (edublitz)
    2. click on browse collections
    3. create 3 databases
        1. products_db
        2. orders_db
        3. users_db
    4. we want the endpoints (credentials)
        1. click on the cluster name (three dot)

    2. click on connect via -> Drivers -> copy the endpoint
5. we want also username & password
    1. click on left panel (on MongoDB logo)
    2. click on database & Network Access
    3. click on Add new database user
    4. click on password authentication
        1. username -----> cbz
        2. password -----> redhat
    5. click on built in role -> add built in role -> atlas admin -> add user

{ add username & password in the DB endpoint url }

To connect database to the server -> Add the IP address of server in MongoDB

1. click on database and network access
2. click on ip access list
3. add ip address
    1. server public-address or 0.0.0.0/0 (anywhere entry)

CONFIGURE BACKEND SERVICES (MICROSERVICES)

USER-SERVICE

1. git clone https://github.com/shubhamkalsait/edublitz-b2b-medical-erp.git
2. cd edublitz-b2b-medical-erp
3. git checkout dev
4. apt install openjdk-17-jdk -y
5. apt install maven -y
6. cd user-service/src/main/resources
7. vim application.yaml
   change the mongodb_url
   mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/users_db?appName=edublitz
8. cd user-service
9. cp .env.example .env
10. vim .env
    update mongodb_url
    mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/users_db?appName=edublitz
11. mvn clean package -DskipTests
12. ls target
13. nohup java -jar target/user-service-1.0.0.jar > user-service.log 2>&1 &
14. lsof -i :8081
15. curl http://localhost:8081/api/v1/actuator/health

PRODUCT-SERVICE

1. cd product-service/src/main/resources
2. vim application.yaml
   change the mongodb_url
   mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/products_db?appName=edublitz
3. cd product-service
4. cp .env.example .env

5. vim .env
   update mongodb_url
   mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/products_db?appName=edublitz
6. mvn clean package -DskipTests
7. ls target
8. nohup java -jar target/product-service-1.0.0.jar > product-service.log 2>&1 &
9. lsof -i :8082
10. curl http://localhost:8082/api/v1/actuator/health

ORDER-SERVICE

1. cd order-service/src/main/resources
2. vim application.yaml
3. change the mongodb_url
   mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/orders_db?appName=edublitz
4. cd order-service
5. cp .env.example .env
6. vim .env
   update mongodb_url
   mongodb+srv://cbz:redhat@edublitz.js8tilw.mongodb.net/orders_db?appName=edublitz
7. mvn clean package -DskipTests
8. ls target
9. nohup java -jar target/order-service-1.0.0.jar > order-service.log 2>&1 &
10. lsof -i :8083
11. curl http://localhost:8083/api/v1/actuator/health

CONFIGURE FRONTEND

1. cd edublitz-b2b-medical-erp/frontend/
2. vim .env.production
   VITE_USER_SERVICE_URL=http://18.218.74.150:8081/api/v1 #add public_ip of backend server
   VITE_PRODUCT_SERVICE_URL=http://18.218.74.150:8082/api/v1
   VITE_ORDER_SERVICE_URL=http://18.218.74.150:8083/api/v1
3. apt install nodejs npm -y
4. npm install
5. npm run build
6. ls dist/
7. apt install nginx -y
8. rm -rf /var/www/html/*
9. cp -rf dist/* /var/www/html/
10. systemctl restart nginx

11. To check the frontend
    <public_ip>
    Login Credentials:
    1. Username: admin@medicalerp.com
    2. Password: Admin@123456

12. To check the backend

1. http://public_ip:8081/api/v1/actuator/health
2. http://public_ip:8081/api/v1/actuator/health
3. http://public_ip:8081/api/v1/actuator/health

DEPLOYING FRONTEND ON S3

1. create a s3 bucket (mayurcbz.space)
   1.bucket should be private
2. create a role and attach to ec2 instance
   1.role ---------> s3FullAccess
3. On server
   1.sudo apt update
   2.sudo snap install aws-cli --classic
   3.aws s3 ls
   4.aws s3 cp dist/. S3://mayurcbz.space --recursive
4. On s3
   1.Go to properties
   2.create a static-website-hosting
   3.create

5.check the URL

ACCESSING APPLICATION DASHBOARD

1.add organisation
   1. add distributor
   2. type ----> distributor
   3. name ---> mayur-co-pharma
   4. registration number ----> abcdef-1234
   5. address
   6. create a distributor

2.add organisation
   1. add hospital
   2. type ----> hospital
   3. name ---> orange city hospital
   4. registration number ----> abc-123
   5. address
   6. create a distributor

“Once both organizations are created, the hospital can purchase medicines from the distributor."

3.Add the products
   1. add products
   2. distributor --- mayur-co-pharma
   3. stock ---------- PARA-123-AUG-26
   4. product name -- PARACETAMOL
   5. Manufacturer -- cipla
   6. category ------ MEDICINE
   7. Unit ---------- strip
   8. product type -- branded

9. MRP ---------- 20
10. Wholesale --- 15
11. create

4.Add the inventory

1. add stock (batch)
2. product ---- PARA-123-AUG-26
3. location ---- Nagpur
4. batch nu --- B2026
5. MFG ------- (optional)
6. Expiry ----- 1/1/2027
7. Quantity - 100
8. reorder alert at --- 10
9. save batch

5.Open new page in incognito mode

<public_ip>

1. register user
2. name ------- mayur
3. email ------- mayur@123.com
4. role --------- hospital
5. org ID ------ <copy & paste the org_id of orange-city-hospital>
6. create account

6.Get the order

1. click on order
2. distributor ---- mayur-co-pharma
3. shipping address --- Nagpur
4. line items ----- PARA-123-AUG-26
5. quantity ------- 20
6. place order

7.Go to Admin Account

1. click on orders
   {you get the status pending}
2. click on these pending
3. approve order
4. mark as dispatched

8.On user

1. confirm delivery
